MILITARY SPECIFICATION

BALANCE, TORSION, 500 gram Capacity

This limited coordination Military specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but it has not been approved for promulgation as a coordinated document. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers a 500-gram capacity weight loader torsion balance for use in laboratories of field medical units.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-1122	Balance, Scale, and Accessories, Packaging and Packing of.

Military

MIL-M-7298	Manual, Technical: Commercial Equipment.

STANDARDS

Federal

FED-STD-66	Steel: Chemical Composition and Hardenability.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6670

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposals shall apply.

MAGNETIC MATERIALS PRODUCERS ASSOCIATION

MMPA 0100-72 Permanent Magnetic Materials.

(Application for copies should be addressed to Magnetic Materials Producers Association, 1717 Howard Street, Evanston, Ill. 60202.)

3. REQUIREMENTS

3.1 Material. Material shall be functionally suitable and as herein specified.

3.2 Construction. Balance shall consist of a base, a weighing mechanism housing, a weighing mechanism, weighing pedestals and pans, weight indicators, and a balancing indicator.

3.2.1 Base. Base shall be diecast and shall be fabricated of zinc. Base shall support weighing mechanism and weighing mechanism housing. A fixed metal leg shall be centrally located beneath the back edge. On the front edge each front corner shall have a threaded hole which mates without binding with threads on metal leveling screws. Leveling screws shall be furnished.

3.2.2 Weighing mechanism. Balancing shall be by means of equal arm beam and torsion bands. Mechanism shall contain no knife edges. A weight loader mechanism shall be attached to the beam. Weight loader shall operate in conjunction with the 0 to 90 dial. Fractions of 10 grams shall be balanced by means of a spring mechanism which operates in conjunction with the 0 to 10 dial. Mechanism shall be damped by a magnetic damper. Magnet shall be in accordance with MMPA 0100-72, Alnico V.

3.2.3 Housing for weighing mechanism. Housing shall be fabricated from cold-rolled steel which is at least 0.04 inch thick. Front face shall contain two dials and a metal arrest and release knob. Dials shall rotate so that they indicate weight in grams and shall be marked with the word "gram" or letter "g". Arrest and release knob shall immobilize the weighing mechanism when turned clockwise as far as possible and shall release the weighing mechanism when turned counterclockwise as far as possible. Arrest and release system shall be fabricated of corrosion-resisting metal. The left side, as determined by viewing the balance from the front, shall have a removable panel. Inside of panel shall contain instructions for placing weights on the weight loader mechanism. Right side shall have a device which indicates whether the back and the front are level. Balancing indicator

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(see 3.2.5) shall be used to level the sides of the balance. Top surface shall have two holes for the pedestals. One hole shall be to the left and one hole shall be to the right of center. Both holes shall be centered between the back and the front.

3.2.4 Weighing pedestals and pan . Pedestals shall have circular and flat tops which are supported by a leg which fits through the hole in the top surface of the housing. Left pedestal shall furnish support for the weighing pan and the right pedestal shall provide space for loose weights. Loose weights are not to be furnished with the balance. Diameter of pedestals shall be 3 ± 1/4 inch. Weighing pan shall be circular and shall be the bowl type. Bottom of pan shall be flat and top perimeter shall be 1/2 ± 1/8 inch above the bottom perimeter. Top diameter shall be 5 ± 1/4 inch. Difference between top and bottom diameter shall be not more than 1/4 inch. Pedestals and weighing pan shall be corrosion-resisting steel in accordance with FED-STD-66, AISI 300 series.

3.2.5 Balancing indicator. Balancing indicator shall consist of 9 lines and a pointer. Indicator shall be located on the top surface of the weighing mechanism housing, near the front edge and centered between the right and left edges. Lines shall be on flat surface, and pointer shall be connected to the balancing portion of the weighing mechanism. Lines shall be perpendicular to the front edge of the weighing mechanism housing. Lines shall be equidistant from each other and the total width covered by the lined area shall be 7/16 ± 1/16 inch. Central line shall be longer than the remainder of the lines and shall indicate the balance and level points. Balancing indicator shall be protected on the sides by metal walls and on the top by a glass panel.

3.2.6 Dials. Front surface of balance shall have one dial which is graduated from 0 to 10 and one dial which is graduated from 0 to 90. For 0 to 10 dial smallest graduated interval shall be 0.1, every fifth graduated marking shall be longer than other graduation markings and every tenth graduation shall be marked with applicable Arabic number. All graduation markings shall extend to the perimeter of the dial and shall indicate weight when aligned with line which shall be marked above the dial. The 0 to 90 dial shall be marked with Arabic numbers in multiples of 10 from 0 to 90 and shall indicate weight when aligned with indicating mark which shall be placed above the dial.

3.3 Overall dimensions. Nominal overall dimensions of the balance without the pedestals in place shall be 11-3/4 x 6 x 6-1/4 inches. Maximum overall dimensions with the balance level and the pedestals and weighing pan in place shall be width 11-3/4 inches, depth 6 inches, and height 7-1/4 inches.

3.4 Weight. Weight of balance shall not exceed 10 pounds.

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3.5 Performance.

3.5.1 Accuracy. Weight on pan shall be identical to dial reading ± 50 milligrams (mg.) when tested as specified in 4.4.1.

3.5.2 Sensitivity requirement (SR). Pointer shall move at least one graduation, when tested as specified in 4.4.2.

3.5.3 Arm ratio. No more than 0.375 grams shall be required to balance weighing pan and weight pedestal.

3.5.4 Shift test. No more than 0.500 grams shall be required to balance weighing pan and weight pedestal.

3.6 Finish. Weighing pan shall be polished to a No. 4 commercial finish. Metals which are not corrosion-resisting shall be plated to resist corrosion except for the weighing mechanism housing and walls of the balance indicator. Corrosive metal portions of the weighing mechanism housing and the balance indicator housing shall be cleaned and sanded, covered with zinc chromate primer and the primer covered with a gray baked-on enamel.

3.7 Identification marking. Each balance shall have a nameplate which shall be permanently and legibly marked with the following information: contract or purchase order number, National Stock Number, capacity and manufacturer's name or registered trademark, and "U.S." Markings secured by adhesive shall not be acceptable.

3.8 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

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3.9 Workmanship. Workmanship shall be first class throughout. Balance shall be free of defects listed in Table III and any other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificates of quality, supplied by the manufacturer of the magnet and steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificates shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I. Unit of product shall be the balance.

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Table I Sampling for examination

	Inspection level	AQL (Percent defective)
For visual examination		
Major A defects	S-2	1.0
Major B defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. Unit of product shall be the balance, and acceptence number shall be zero.

Table II Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level
Accuracy	3.5.1	4.4.1	S-1
Sensitivity	3.5.2	4.4.2	100%
Arm ratio	3.5.3	4.4.3	S-1
Shift	3.5.4	4.4.4	S-1

4.3 Examination. Balance shall be examined to determine compliance with all requirements contained in this specification. The balance shall be examined in accordance with Table III, but examination shall not be limited to defects listed therein.

Table III Classification of defects

Category	Defect *
Major A	
101	Balance does not function when service data and instructions on balance are followed.
Major B	
151	Balance not free of sharp edges.
152	Surface of weighing pans not smooth in all areas.
153	Component missing.
Minor	
201	Plated and enameled surfaces not smooth.
202	Corrosion-resisting metal surfaces not smooth.
203	Plating not continuous.
204	Enameling not continuous.
205	Enamel not free of blisters.
206	Identification marking not complete, correct, legible or permanent.

* Examination shall not be restricted to the defects listed above.

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4.3.1 Dimensional examination. Balance shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accuracy. Balance shall be leveled in accordance with instructions in service data and on the balance. The 0 to 90 gram dial shall be set at 90 grams and the pointer on the balancing indicator shall be centered by placing test weights on the weighing pan. Differential between weights and dial reading shall be recorded. The 0 to 10 gram dial shall be set at 10 grams and pointer on the balancing indicator shall be centered by adding test weights to the weighing pan. Differential between the added weights and dial reading shall be recorded. Differential of more than 50 mg. for either dial shall be considered cause for rejection. In addition, differentials shall be added arithmetically; aggregate error of more than 50 mg. shall be considered cause for rejection.

Examples

	Weight in grams	Differential in grams
90 gram dial	90.0049	+ .049
10 gram dial	10.0049	+ .049
	Aggregate error	.098
90 gram dial	90.0049	+ .049
10 gram dial	9.9951	- .049
	Aggregate error	.000

Test weights shall be of known value. Any tolerance on test weights shall be subtracted from the tolerance specified.

4.4.2 Sensitivity. Balance shall be leveled in accordance with instructions in service data and on the balance. Arrest and release knob shall be left in the release position. A 50-mg. weight shall be placed on the weighing pan.

4.4.3 Arm ratio. Unit shall be balanced with no load on weighing pan or weight pedestal. Weighing system (weighing mechanism plus weighing pan and weight pedestal) shall be immobilized. Place one 200-gram weight on the pan and one 200-gram weight on the pedestal. Weights shall be placed in the center of the pan and pedestal. Weighing system shall be released and unit shall be balanced by the addition of weights to the pan or pedestal.

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4.4.4 Shift test. Unit shall be balanced with no load on weighing pan or weight pedestal. Weighing system shall be immobilized. One 200-gram weight shall be placed in the center of the pedestal and one 200-gram weight shall be placed equidistant between the center and the front edge of the pan. Weighing system shall be released. Unit shall be balanced by the addition of weights to the pan or pedestal. This procedure shall be repeated 7 times by placing the 200--gram weight equidistant between the center and the left, back and right edges of the pan and by centering the 200--gram weight on the pan and by placing the other 200-gram weight equidistant between the center and the front, left, back and right edges of the pedestal

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking for Levels A, B or C shall be in accordance with PPP-B-1122.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2. This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type normally procured to meet Military requirements.

6.3 This specification covers the following item listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6670-00-022-8013	BALANCE, TORSION, 500 gram Capacity

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6670-0120

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-B-36836 (DSA-DM)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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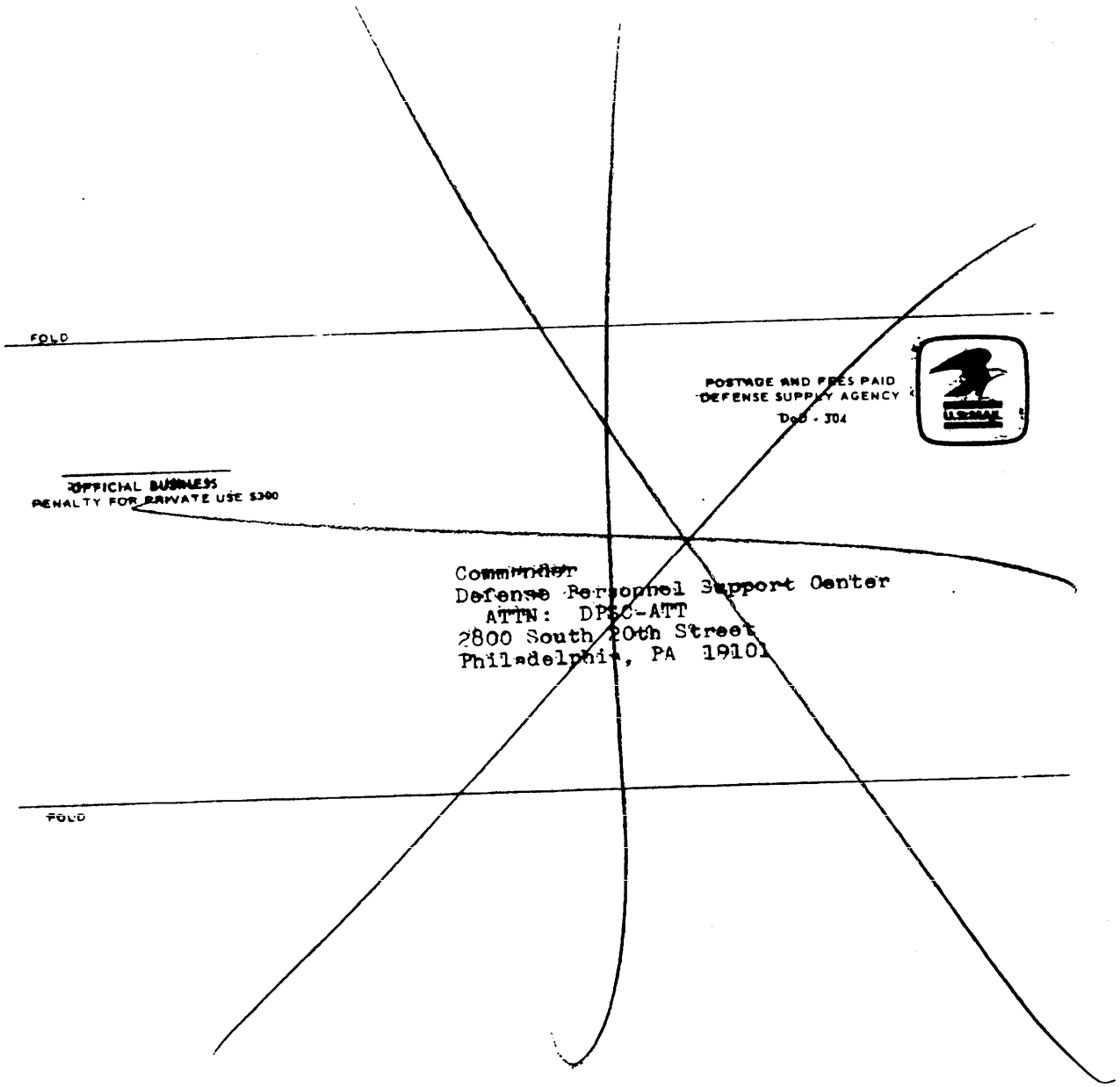
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